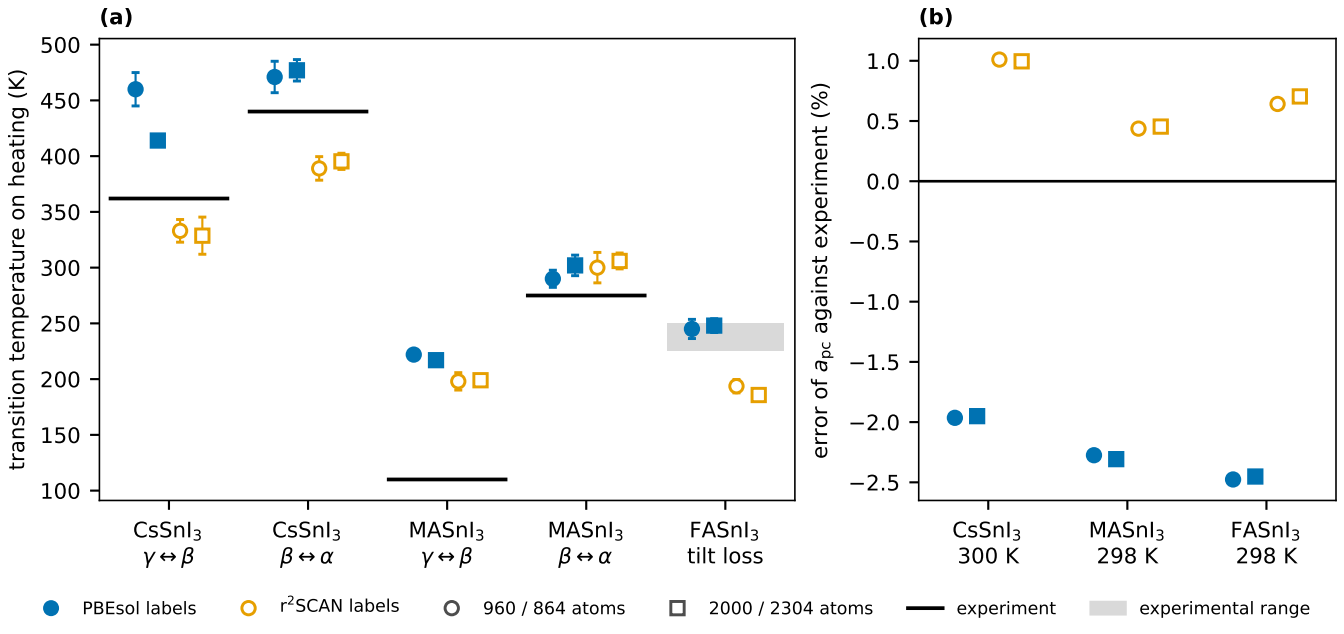
(a)
transition temperature on heating (K)
500
450
400
350
300
250
200
150
100
CsSnI3 γ ↔ β
CsSnI3 β ↔ α
MASnI3 γ ↔ β
MASnI3 β ↔ α
FASnI3 tilt loss
(b)
error of apc against experiment (%)
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
CsSnI3 300 K
MASnI3 298 K
FASnI3 298 K
PBEsol labels
r2SCAN labels
960 / 864 atoms
2000 / 2304 atoms
experiment
experimental range